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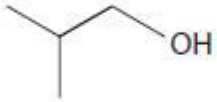
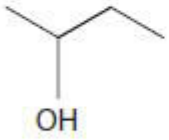
Questions

Q1.

Biobutanol is a possible alternative to bioethanol as an additive to petrol.

(a) (i) Complete the table for the four isomers of C_4H_9OH which are alcohols.

(3)

Skeletal formula	Name	Classification	Boiling temperature / °C
	butan-1-ol	primary	117
			108
			99
	2-methylpropan-2-ol		82

(ii) Explain the difference in boiling temperature between butan-1-ol and 2-methylpropan-2-ol.

(2)

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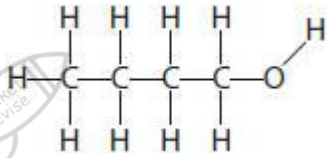
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(iii) The displayed formula of butan-1-ol is shown.
Complete the diagram to show the strongest intermolecular force between **two** molecules of butan-1-ol. Include the intermolecular bond angle.

(2)

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(b) (i) Complete the equation for the combustion of butan-1-ol.

State symbols are not required.

(1)



(ii) Calculate a value for the enthalpy change of combustion of butan-1-ol using the equation in (b)(i) and the mean bond enthalpies in the table.

(3)

Bond	C—C	C—O	C=O	C—H	O—H	O=O
Mean bond enthalpy / kJ mol^{-1}	347	358	805	413	464	498

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(iii) A data book value of the molar enthalpy change of combustion of butan-1-ol is −2670 kJ mol^{-1}
Give **two** reasons for the difference between this value and the value calculated in (b)(ii).

(2)

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(c) Biobutanol has some advantages over bioethanol.

(i) The combustion of bioethanol releases 23 MJ dm^{-3} compared to petrol which releases 32 MJ dm^{-3}. Calculate the energy released in MJ dm^{-3} for the combustion of biobutanol.

Biofuel	Formula	$\Delta_c H$ / kJ mol^{-1}	Density / g cm^{-3}
biobutanol	C_4H_9OH	−2670	0.810

(2)

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(ii) Biobutanol can be mixed with petrol in any proportion whereas bioethanol cannot. Petrol is a mixture of liquid alkanes.
Explain why petrol is more miscible with biobutanol than with bioethanol.

(2)

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(Total for question = 17 marks)

Q2.

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Iodine can be extracted from seaweed. The seaweed is heated strongly to burn off the organic matter. The resultant ash is boiled in water to dissolve the iodide ions, and the mixture is filtered.

(a) Acidified hydrogen peroxide (H_2O_2) is added to the filtrate.

A redox reaction takes place with iodine and water as the only products.

(i) Write half-equations for the oxidation and reduction reactions that take place. State symbols are not required.

(2)

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(ii) Use your answers to (a)(i) to write the overall equation for this redox reaction.
State symbols are not required.

(1)

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(b) The iodine is separated from the aqueous solution using solvent extraction.

The aqueous solution is mixed with cyclohexane in a separating funnel, forming two layers.
The mixture is then shaken gently and left until the layers separate.
Most of the iodine dissolves in the cyclohexane layer.

(i) State the colour of each layer **after** separation.

(2)

Aqueous layer .....

Cyclohexane layer
.....

(ii) Explain why iodine is very soluble in cyclohexane but only slightly soluble in water.

(2)

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(c) The cyclohexane layer is then removed from the separating funnel and dried.

Identify, by name or formula, a suitable drying agent.

(1)

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(Total for question = 8 marks)

Q3.

A chloroalkane is heated with dilute aqueous sodium hydroxide and the pure organic product is obtained.

When the organic product is warmed with acidified potassium dichromate(VI) solution, there is no change in colour.

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The chloroalkane could be

(1)

- ☒ **A** 1-chlorobutane
- ☒ **B** 2-chloro-2-methylpropane
- ☒ **C** 1-chloro-2-methylpropane
- ☒ **D** 2-chlorobutane

(Total for question = 1 mark)

Q4.

Propanoic acid, CH_3CH_2COOH, is a colourless liquid used as a preservative in animal feed.
Propanoic acid can be formed by oxidising the alcohol propan-1-ol.

(a) Write the balanced equation for the oxidation of propan-1-ol to form propanoic acid.

Use [O] to represent the oxygen from the oxidising agent.
State symbols are not required.

(1)

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*(b) Propan-1-ol is heated with a concentrated solution of acidified potassium dichromate(VI).

Explain how the conditions used affect the rate of the reaction **and** ensure that propanoic acid is the only organic product.

(6)

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(c) A student suggested using universal indicator to check for the presence of propanoic acid formed in the reaction mixture.
Give a reason why the result of this test is **not** likely to be conclusive.

(1)

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(d) The permitted mass of propanoic acid used in animal feed is in the range 1000–3000 mg kg^{-1}. A titration method may be used to check the concentration of propanoic acid in animal feed.
A 50.0 cm^3 sample of propanoic acid solution was extracted from 50 g of an animal feed.
The sample was added to a volumetric flask and the volume made up to 250.0 cm^3 and mixed thoroughly. A pipette was used to transfer 25.0 cm^3 of the diluted acid into a conical flask containing an indicator.
The contents of the conical flask were titrated with a solution of sodium hydroxide, NaOH(aq), with concentration 0.00668 mol dm^{-3}.
The procedure was repeated twice and the results obtained are shown.

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	Run 1	Run 2	Run 3
Titre / cm^3	23.20	22.10	22.20

(i) Phenolphthalein is a suitable indicator for this titration.
State the colour **change** at the end-point.

(2)

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(ii) Suggest **two** possible reasons why the titre for Run 1 is greater than the other two titres.

(2)

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(iii) The equation for the reaction of propanoic acid with sodium hydroxide is

$$CH_3CH_2COOH + NaOH \rightarrow CH_3CH_2COONa + H_2O$$

Calculate the mass in grams of propanoic acid extracted from the animal feed.
Give your answer to an appropriate number of significant figures.

(5)

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(iv) Use your answer to (d)(iii) to determine whether the acid in this sample lies within the permitted range for use in animal feed.

(2)

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(v) Suggest how the animal feed would be affected if the amount of propanoic acid was outside the permitted range.

(1)

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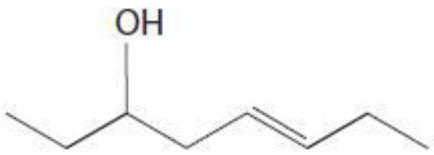
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(Total for question = 20 marks)

Q5.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

(a) What is the name of compound L?



L

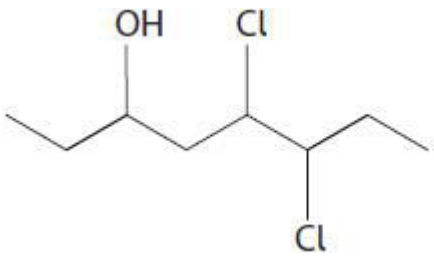
(1)

- ☒ **A** *Z*-oct-3-en-3-ol
- ☒ **B** *E*-oct-3-en-3-ol
- ☒ **C** *Z*-oct-5-en-3-ol
- ☒ **D** *E*-oct-5-en-3-ol

(b) Compound L can be converted into compound M.

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M

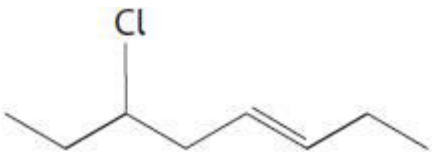
Which reagent should be used?

(1)

- **A** $Cl_2(g)$
- **B** HCl(g)
- **C** $PCl_5(s)$
- **D** KCl(aq)

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(c) Compound L can also be converted into compound N.



N

Which reagent should be used?

(1)

- **A** $Cl_2(g)$
- **B** HCl(g)
- **C** $PCl_5(s)$
- **D** KCl(aq)

(Total for question = 3 marks)

Q6.

1-bromopropane is used for spot removal when 'dry cleaning' clothes.

(a) 1-bromopropane does not occur naturally but can be made from propan-1-ol.

Identify the reagent or reagents you would use to make 1-bromopropane from propan-1-ol.

(1)

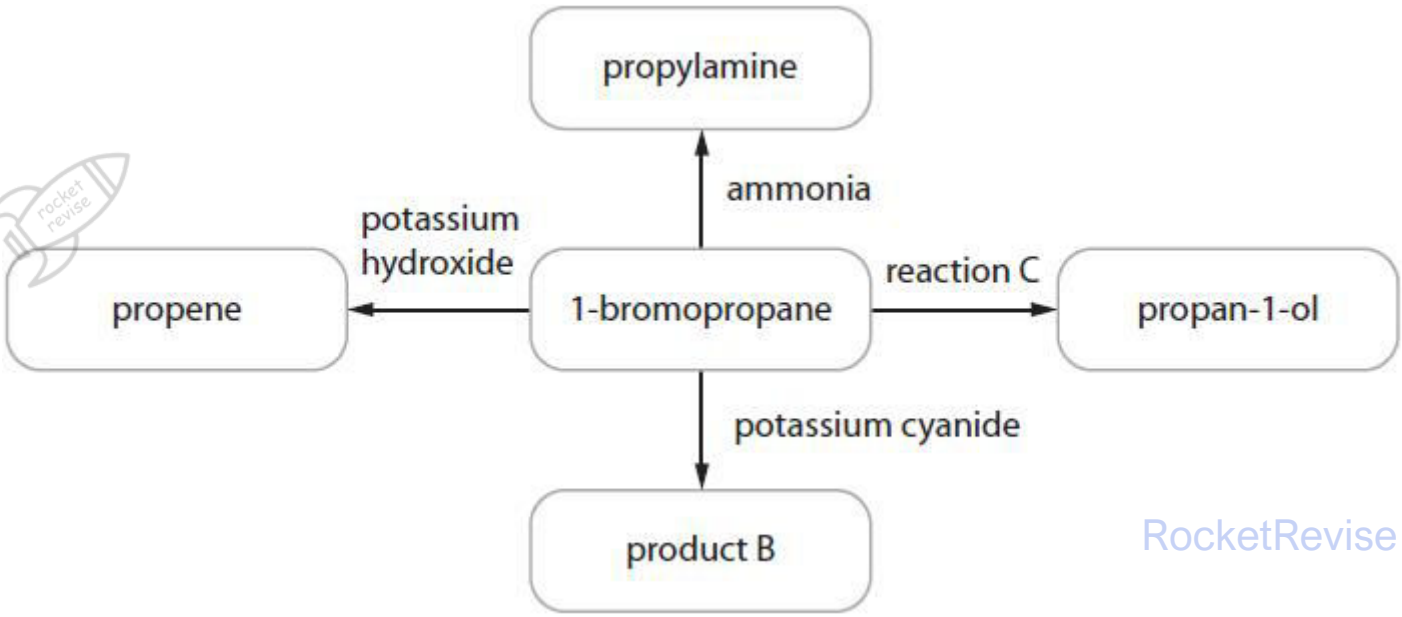
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(b) Some reactions of 1-bromopropane are shown.

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(i) Give the conditions for the formation of propene.

(1)

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(ii) Give the **molecular** formula of product B.

(1)

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(iii) Name the type and mechanism of the reaction taking place in reaction C.

(2)

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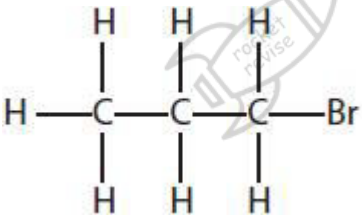
(iv) Complete the mechanism for the reaction that occurs between ammonia and 1-bromopropane to form propylamine,

$CH_3CH_2CH_2NH_2$.

Include curly arrows, and relevant lone pairs and dipoles.

(4)

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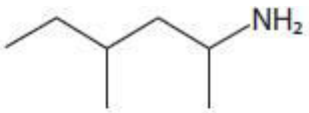
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(Total for question = 9 marks)

Q7.

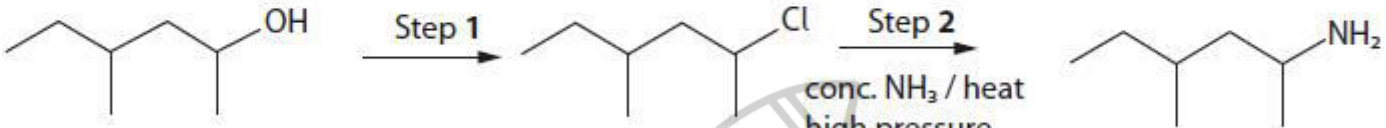
The compound DMAA was originally synthesised as a decongestant.



DMAA

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A suggested synthetic route for DMAA is shown.



compound **A**

(a) (i) Give the systematic name of compound **A**.

(1)

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(ii) Identify, by name or formula, a suitable reagent for Step **1**.

(1)

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(iii) Give the mechanism for the reaction in Step **2**.
Include curly arrows, and any relevant dipoles and lone pairs.

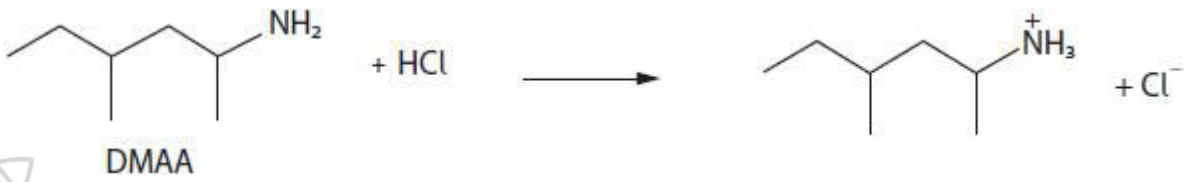
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(b) DMAA is only slightly soluble in water but dissolves readily in hydrochloric acid to form an aqueous solution.

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DMAA

(i) Explain the type of bonding that occurs between the nitrogen atom and the hydrogen ion, when the positive ion forms.

(3)

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(ii) Complete the diagrams to show how the ions formed in the reaction between DMAA and hydrochloric acid interact with water molecules.

(3)



(Total for question = 12 marks)

Q8.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Some alcohols react with concentrated phosphoric(V) acid, H_3PO_4, to form alkenes.

(a) What type of reaction occurs?

☐ **A** addition

☐ **B** elimination

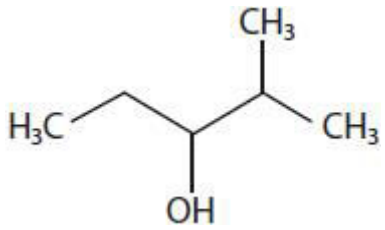
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☒ **C** hydrolysis

☒ **D** substitution

(b) The structure of 2-methylpentan-3-ol is shown.



This alcohol reacts with concentrated phosphoric(V) acid.
How many different alkenes can form?

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☒ **A** one
☒ **B** two
☒ **C** three
☒ **D** four

(Total for question = 2 marks)

Q9.

Ethanol can be made in industry by the reaction of ethene with steam, using a phosphoric(V) acid catalyst.

$CH_2{=}CH_2(g) + H_2O(g) \rightleftharpoons CH_3CH_2OH(g) \quad \Delta_rH = -45 \text{ kJ mol}^{-1}$

The reaction is carried out at 300°C and 60 atm. An initial yield of 5 % is achieved when the ethene and steam first pass through the reactor.

*(a) Explain the chemical reasons for the conditions used and why such a low initial yield is acceptable in the industrial process.

(6)

(b) The product of the reaction is a mixture of ethanol and water.

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Explain why ethanol and water mix together fully.
You may find it helpful to draw a diagram.

(3)

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(c) Ethanol can be oxidised using a solution of acidified potassium dichromate(VI), $K_2Cr_2O_7(aq)$.

Ethanoic acid and another organic compound, **Y**, are both possible products.

(i) Draw the structure of **Y**.

(1)

(ii) State the conditions needed to maximise the yield of each product.

(2)

Conditions for maximum yield of **Y**

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Conditions for maximum yield of ethanoic acid

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(Total for question = 12 marks)

Q10.

This question is about ethanol and bioethanol.

The main fuel used as a petrol substitute is bioethanol. Bioethanol is ethanol that has been produced by fermentation. The starting material is usually some form of plant material rich in starch, such as wheat, maize or potatoes. Enzymes in yeast convert this material to simple carbohydrates such as glucose ($C_6H_{12}O_6$) and then to ethanol and carbon dioxide.

$$C_6H_{12}O_6 \rightarrow 2CH_3CH_2OH + 2CO_2$$

The mixture is left for several days until fermentation is complete.
The percentage of ethanol is never greater than 15 % because higher concentrations of ethanol kill the yeast.

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A common blend of fuel is 95 % petrol and 5 % bioethanol. The engine does not need to be modified for this mixture.

(a) Give **one** advantage and **one** disadvantage of using bioethanol in petrol.

(2)

Advantage

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Disadvantage

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(b) Suggest why this fermentation must be carried out in the absence of air.

(1)

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(c) Suggest how the ethanol can be obtained, after filtering the fermentation mixture.

(1)

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(d) Ethanol is hygroscopic, which means it readily absorbs water from the air.

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(i) Give a possible reason why ethanol is able to absorb water.

(1)

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(ii) Suggest a problem arising from the hygroscopic nature of ethanol when using this fuel in a motor vehicle.

(1)

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(e) Ethanol can also be produced by the hydration of ethene.

$$CH_2 = CH_2(g) + H_2O(g) \leftrightarrow CH_3CH_2OH(g) \qquad \Delta H = -45 \text{ kJ mol}^{-1}$$

*(i) Typical conditions are 300 °C and 60 atm with a catalyst of phosphoric acid.
Explain why these conditions are used, by describing the effect of changing the temperature and pressure on rate of reaction, equilibrium yield and cost.

(6)

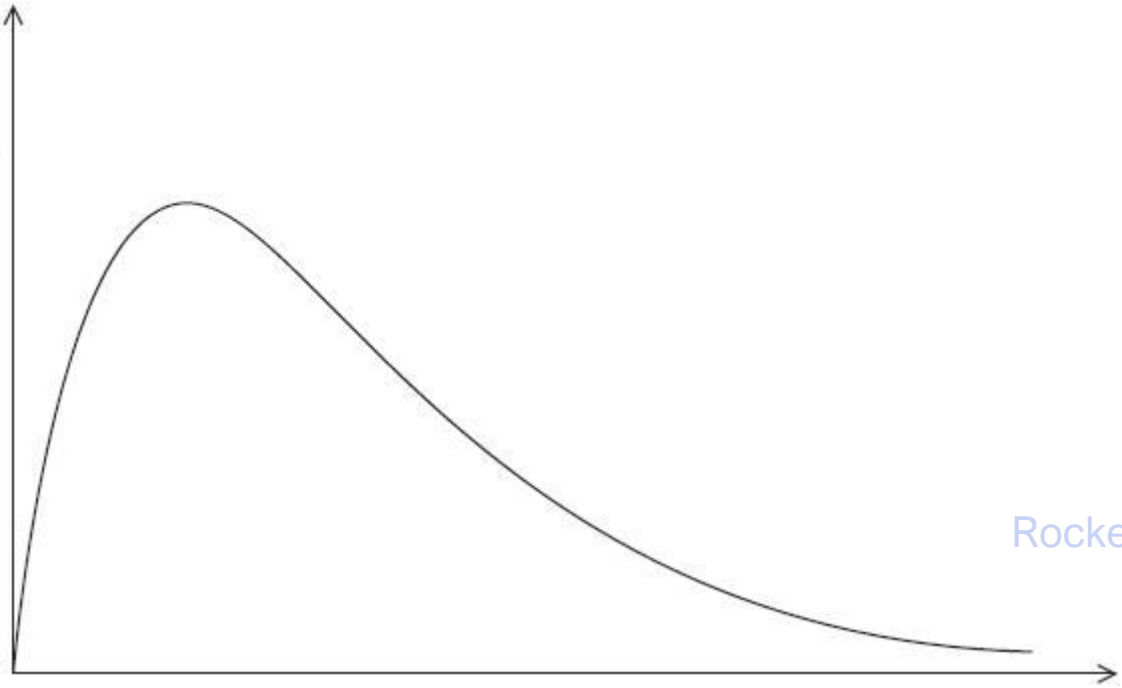
(ii) The rate of this reaction is increased by using a catalyst of phosphoric acid.
Label the axes on the Maxwell–Boltzmann distribution curve and use it to explain how a catalyst increases the rate of reaction.

(4)

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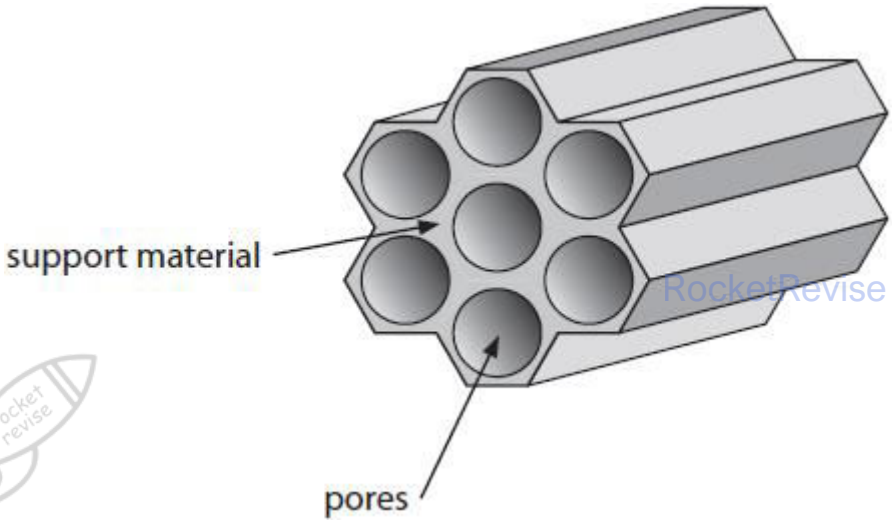
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(f) Catalysts such as phosphoric acid are bonded to a support material that contains lots of pores.



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(i) Suggest the advantage of using support materials containing lots of pores.

(1)

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(ii) Under these conditions, only about 5 % of the ethene is converted into ethanol as it passes over the catalyst.
Suggest how the overall yield of this process can be improved to make it economically viable.

(2)

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(Total for question = 19 marks)

Q11.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

How many moles are there in 15.1 cm^3 of liquid propan-1-ol?

[Density of propan-1-ol = 0.80 g cm^{-3} M_r of propan-1-ol = 60]

☒ **A** (0.80 × 15.1) ÷ 60

☒ **B** 0.80 ÷ (60 × 15.1)

☒ **C** 60 ÷ (0.80 × 15.1)

☒ **D** (60 × 15.1) ÷ 0.80

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(Total for question = 1 mark)

Q12.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The oxidation of propan-1-ol by acidified potassium dichromate(VI) forms propanoic acid with a yield of 36 % by mass.

What mass of propan-1-ol is needed to form 37.0 g of propanoic acid in this reaction?

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[M_r of propanoic acid = 74 M_r of propan-1-ol = 60]

☒ **A** 10.8 g

☒ **B** 36.0 g

☒ **C** 40.8 g

☒ **D** 83.3 g

(Total for question = 1 mark)

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Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these compounds does **not** react with acidified potassium dichromate(VI)?

☒ **A** CH_3CH_2OH

☒ **B** $CH_3CHOHCH_3$

☒ **C** CH_3CH_2CHO

☒ **D** CH_3COCH_3

(Total for question = 1 mark)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these compounds is a tertiary alcohol?

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☒ **A** 2-methylpropan-2-ol

☒ **B** 3-methylbutan-2-ol

☒ **C** 2,2-dimethylpropan-1-ol

☒ **D** 3,3-dimethylbutan-2-ol

(Total for question = 1 mark)

Q15.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

How many structural isomers with the molecular formula $C_5H_{10}O$ react with Benedict's or Fehling's solutions?

- ☐ **A** 2
- ☐ **B** 3
- ☐ **C** 4
- ☐ **D** 5

(Total for question = 1 mark)

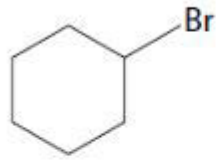
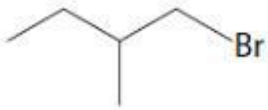
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Q16.

This question is about some halogenoalkanes.

(a) Give the classification of each halogenoalkane shown in the table as primary, secondary or tertiary.

(1)

Halogenoalkane	Classification
	
	

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(b) Samples of 1-iodobutane and 2-chloro-2-methylpropane were hydrolysed using aqueous silver nitrate in ethanol.

Explain whether or not it is possible to predict the relative rate of hydrolysis of these two compounds.

(3)

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(c) Butylamine is formed when 1-bromobutane reacts with excess concentrated alcoholic ammonia.

$$CH_3CH_2CH_2CH_2Br + 2NH_3 \rightarrow CH_3CH_2CH_2CH_2NH_2 + NH_4Br$$

(i) Give a reason why this reaction should be carried out by heating the reactants in a sealed tube rather than heating under reflux.

(1)

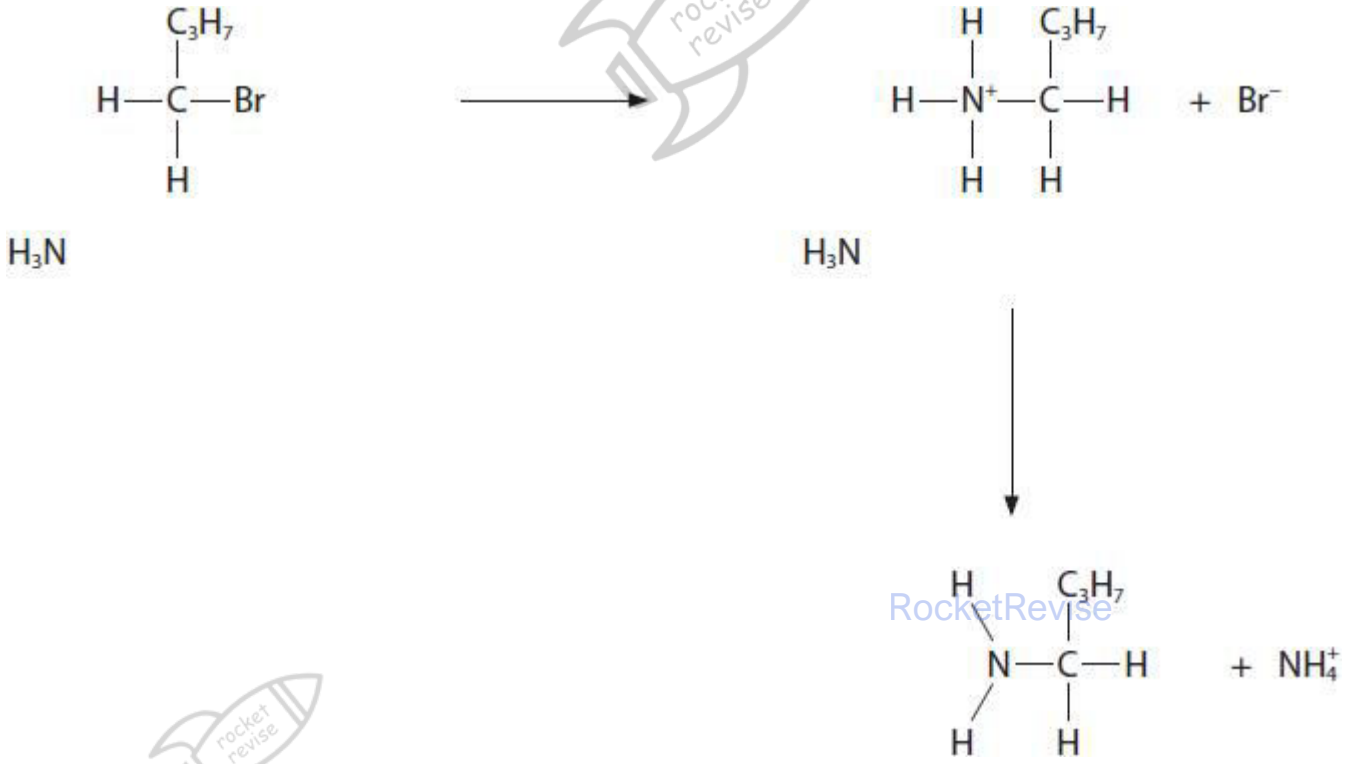
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(ii) Complete the mechanism for this reaction by adding curly arrows, and any relevant lone pairs and dipoles.

(4)



(d) Bromoethane is prepared by reacting ethanol with potassium bromide and concentrated sulfuric acid.

$$C_2H_5OH + KBr + H_2SO_4 \rightarrow C_2H_5Br + KHSO_4 + H_2O$$

Calculate the maximum mass of bromoethane that could be prepared from 4.65 g of ethanol, 14.90 g of potassium bromide and a solution containing 16.35 g of H_2SO_4.

[Ar values: H = 1.0 C = 12.0 O = 16.0 S = 32.1 K = 39.1 Br = 79.9]

(3)

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*(e) The reaction between 2-bromopropane and potassium hydroxide takes place under two different conditions:

- in aqueous solution
- in ethanolic solution.

Compare and contrast these two reactions.
Include equations for the reactions.
Detailed mechanisms of these reactions are **not** required.

(6)

(Total for question = 18 marks)

Q17.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which reagent would convert an alcohol into an alkene?

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- ☒ **A** acidified potassium dichromate(VI)
- ☒ **B** anhydrous calcium sulfate
- ☒ **C** concentrated phosphoric acid
- ☒ **D** ethanolic potassium hydroxide

(Total for question = 1 mark)

Q18.

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This question is about bicycles.

Bicycle frames have been made from steel and aluminium for many years but more recently titanium has been used.

Titanium is a metal that is extracted from its ore, impure titanium oxide (TiO_2), in a two-step process.

Step **1**

The titanium oxide reacts with chlorine in the presence of carbon to form titanium(IV) chloride ($TiCl_4$) and carbon monoxide.

Step **2**

The titanium(IV) chloride reacts with magnesium.

$$TiCl_4 + 2Mg \rightarrow Ti + 2MgCl_2$$

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(a) (i) Write the equation for Step **1**.

State symbols are not required.

(1)

(ii) Explain why Step **2** is a redox reaction.
Refer to relevant oxidation numbers in your answer.

(2)

.....

.....

.....

(b) Titanium(IV) chloride is a liquid that reacts violently with water, producing white smoke. This reaction has been used to produce naval smoke screens to hide ships.

Suggest the type of reaction taking place **and** the compound producing the white smoke.

(2)

Type of reaction

.....

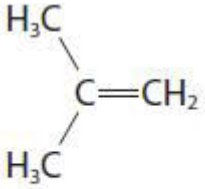
Compound

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.....

(c) Many bicycles use an inner tube inside the tyre.

A common material used for inner tubes is the addition polymer formed from 2-methylpropene.



2-methylpropene

Draw a section of the polymer showing two repeat units.

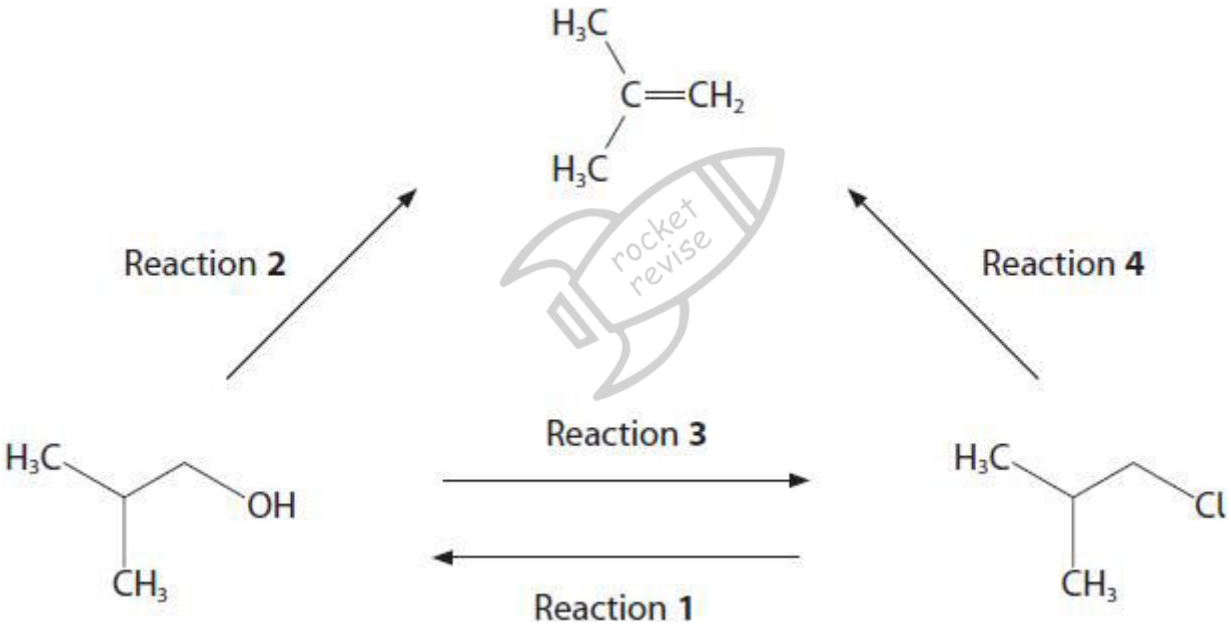
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(2)

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(d) A reaction scheme involving 2-methylpropene is shown.



(i) For Reaction **1**, give the name or formula of a suitable reagent, stating the conditions for the reaction.

(2)

Reagent

.....

Conditions

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.....

(ii) For Reaction **2**, give the name or formula of a suitable reagent, stating the type of reaction taking place.

(2)

Reagent

.....

Type of reaction

.....

(iii) For Reaction **3**, give the name or formula of a suitable reagent, writing an equation for the reaction taking place.

(2)

Reagent

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.....

Equation

(iv) For Reaction **4**, the reagent used is potassium hydroxide (KOH).
Give the conditions required for this reaction, stating the role of the hydroxide ions in this reaction.

(2)

Conditions

.....

Role of hydroxide ions

.....

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(e) Many cyclists carry a small cylinder of carbon dioxide to inflate their inner tube if it is punctured. The carbon dioxide is stored under pressure.



(Source: © Douglas Sacha/Getty Images)

A gas cylinder contains 16.0 g of carbon dioxide (CO_2).
The cylinder has a volume of 20 cm^3 .
Calculate the pressure in the cylinder at 25 °C. Include units in your answer.
Assume that all the CO_2 in the cylinder is a gas.
[$pV = nRT$ $R = 8.31$ J mol^{-1} K^{-1}]

(5)

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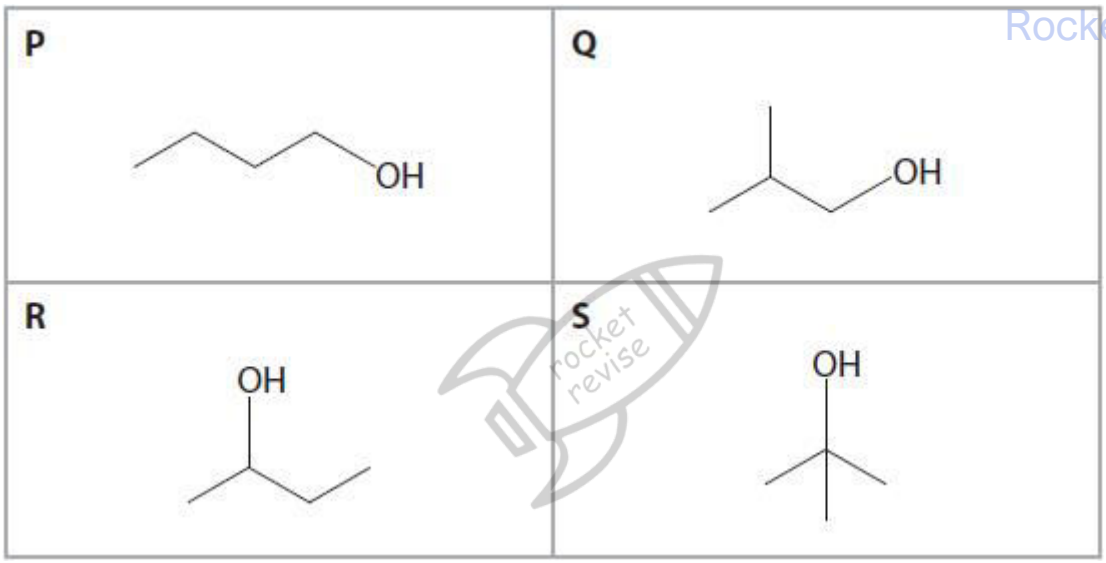
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(Total for question = 20 marks)

Q19.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these alcohols will produce a carboxylic acid when heated under reflux with acidified potassium dichromate(VI)?



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- ☐ **A** **P** only
- ☐ **B** **P** and **Q** only
- ☐ **C** **P**, **Q** and **R** only
- ☐ **D** **P**, **Q**, **R** and **S**

(Total for question = 1 mark)

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Q20.

Ammonium nitrate, NH_4NO_3, is used in the manufacture of fertilisers and explosives.
It is produced on a large scale using only methane, water and air.
The process has four stages.

(a) The first two reactions in Stage **1** involve the production of hydrogen.

At temperature T_1, methane reacts with excess steam to give hydrogen.

$$CH_4(g) + H_2O(g) \rightleftharpoons 3H_2(g) + CO(g) \qquad \Delta H = +206\,kJ\,mol^{-1}$$

At a different temperature, T_2, the carbon monoxide reacts with more steam.

$$CO(g) + H_2O(g) \rightleftharpoons H_2(g) + CO_2(g) \qquad \Delta H = -42\,kJ\,mol^{-1}$$

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(i) Give the reason why excess steam is used in the first reaction.

(1)

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(ii) Predict which of T_1 and T_2 is the **higher** temperature.
Justify your answer.

(1)

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(iii) Derive the **overall** equation for the production of H_2 in Stage **1**.

(1)

(b) The third reaction in Stage **1** involves the removal of carbon dioxide, using an aqueous solution of N-methyldiethanolamine, $CH_3N(CH_2CH_2OH)_2$.

$$CO_2(g) + H_2O(l) + 2CH_3N(CH_2CH_2OH)_2(aq) \rightleftharpoons (CH_3NH(CH_2CH_2OH)_2)_2CO_3(aq)$$

(i) Suggest **one** reason why CO_2 is removed.

(1)

.....

.....

(ii) Name the type of reaction occurring.

(1)

.....

(iii) Draw the **displayed** formula of N-methyldiethanolamine, $CH_3N(CH_2CH_2OH)_2$.

(1)

(c) In Stage **2**, the hydrogen from Stage **1** reacts with nitrogen (from the air) to produce ammonia. The conditions for this reaction are:

- a temperature of 700 K
- a pressure in the range 100–200 atm

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- an iron catalyst

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92\,kJ\,mol^{-1}$$

Give **one** advantage and **one** disadvantage of using a pressure of 200 atm, compared to a pressure of 100 atm, in Stage **2**.

(2)

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(d) The reaction in Stage **2** has an activation energy, E_{cat} = +70 kJ mol^{-1} .

The **uncatalysed** reaction between N_2 and H_2 has an activation energy, E_a = +290 kJ mol^{-1}.

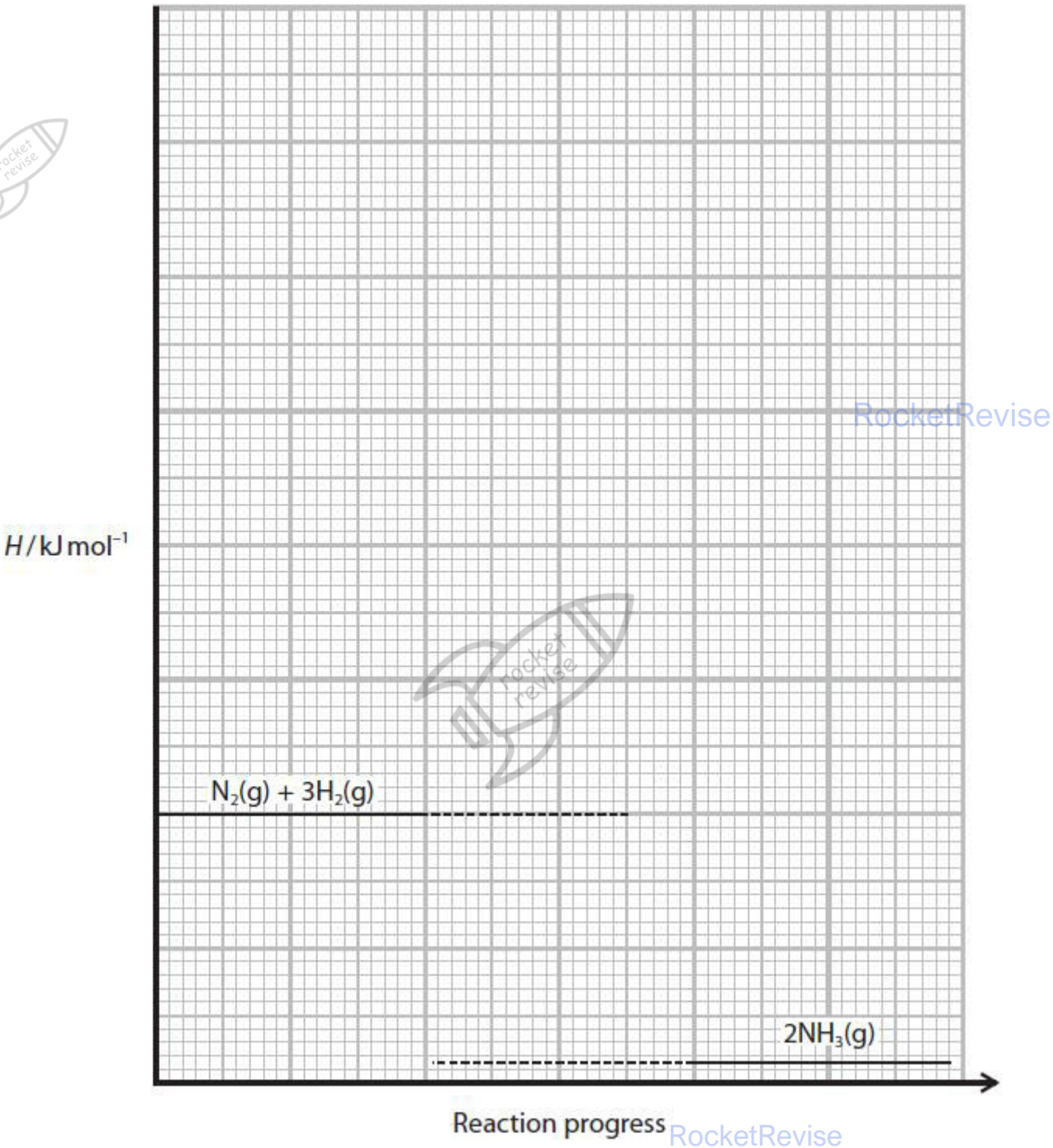
(i) Complete the profile for the catalysed and uncatalysed reactions.
Label the activation energies and the enthalpy change of reaction, ΔH.
Your diagram **must** match the scale shown for the production of NH_3.

(3)

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(ii) Suggest why the use of the catalyst makes Stage **2** more sustainable.

(1)

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(e) In Stage **3**, nitrogen monoxide, NO, is produced in the reaction between NH_3 (from Stage **2**) and O_2 (from the air). The conditions used are a temperature of 1100 K in the presence of a platinum-rhodium catalyst.

$$4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g) \qquad \Delta H = -905\ kJ\,mol^{-1}$$

(i) Give **one** reason why a high temperature is needed in this reaction.

(1)

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(ii) Suggest why only a small amount of energy is used to maintain the temperature at 1100 K.

(1)

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(f) The NO from the first reaction in Stage **3** is cooled and then converted to nitrogen dioxide, NO_2, by reaction with more O_2.

$$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$$

Nitric acid, $HNO_3(aq)$, is produced by the addition of water.

$$3NO_2(g) + H_2O(l) \rightarrow 2HNO_3(aq) + NO(g)$$

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Explain how adding water in the second reaction affects the yield of NO_2 in the first reaction.

(2)

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(g) In Stage **4**, a solution of NH_4NO_3 is produced by reacting NH_3 (from Stage **2**) with HNO_3 (from Stage **3**).

$$NH_3(g) + HNO_3(aq) \rightarrow NH_4NO_3(aq)$$

Data

Species	$NH_3(g)$	$HNO_3(l)$	$NH_4NO_3(s)$
$\Delta_f H^{\ominus}$ / kJ mol^{-1}	−46.1	−174.1	−365.6

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Equation	ΔH / kJ mol^{-1}
$HNO_3(l) + aq \rightarrow HNO_3(aq)$	−32.6
$NH_4NO_3(s) + aq \rightarrow NH_4NO_3(aq)$	+25.6

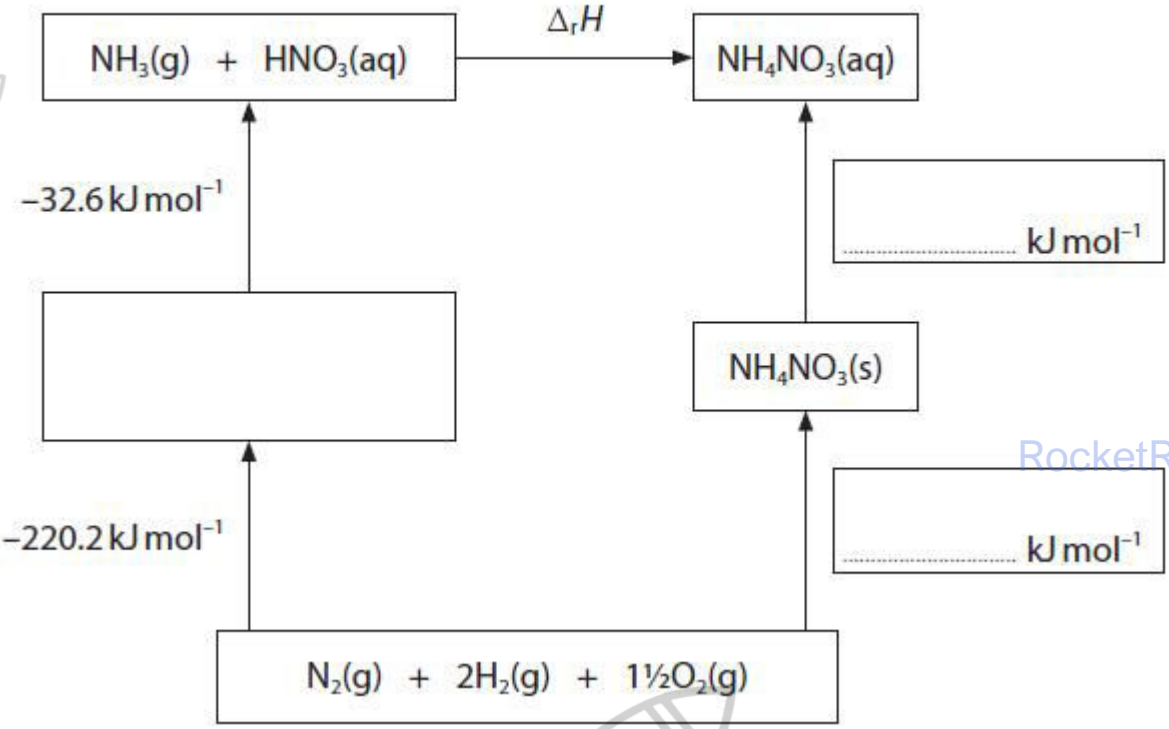
(i) Complete the enthalpy cycle.

(2)

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Enthalpy cycle



(ii) Calculate the enthalpy change, $\Delta_r H$, in kJ mol^{-1}, for the reaction of $NH_3(g)$ with $HNO_3(aq)$.

(1)

(h) Suggest **two** reasons why it is more profitable to carry out all four stages at the **same** site, instead of using different sites for each stage in the industrial production of ammonium nitrate.

(2)

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(Total for question = 21 marks)

Q21.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

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A sample of propane-1,2-diol with a mass of 1.52 g reacts completely with excess phosphorus(V) chloride.

What is the maximum mass, in grams, of the organic product?

[A_r values: H = 1.0 C = 12.0 O = 16.0 Cl = 35.5]

- **A** 1.13
- **B** 1.89
- **C** 2.26
- **D** 4.52

(Total for question = 1 mark)

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Q22.

This question is about alcohols with the molecular formula $C_6H_{14}O$.

(a) Draw the **skeletal** formula of each of the three **tertiary** alcohols with the formula $C_6H_{14}O$.

(3)

tertiary alcohol 1

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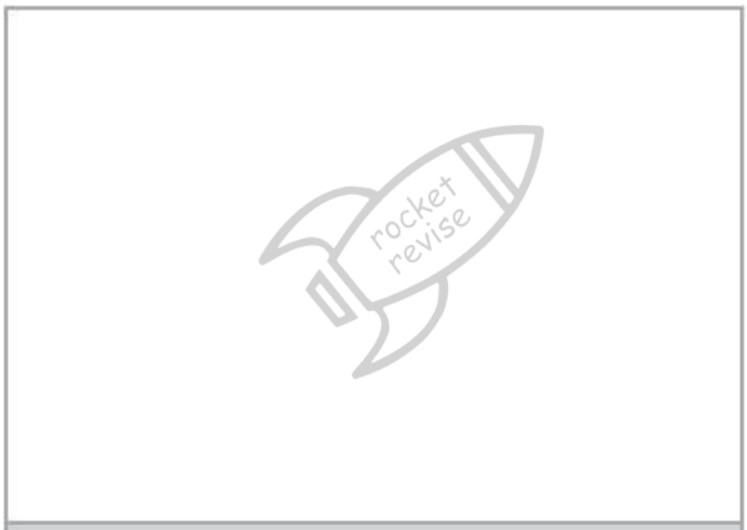
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tertiary alcohol 2

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tertiary alcohol 3

(b) Two primary $C_6H_{14}O$ alcohols, **A** and **B**, are shown.

(i) Give the IUPAC name of alcohol **A**.

(1)

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(ii) Explain why alcohol **B** has a higher boiling temperature than alcohol **A**.

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(2)

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(iii) Explain why alcohol **B** is completely soluble in ethanol but only slightly soluble in water.
A detailed description of the forces involved is **not** required.

(3)

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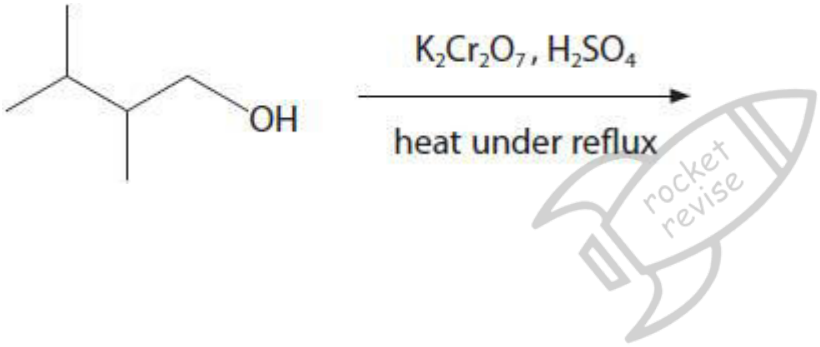
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(c) Give the structure of the **organic** product of each reaction shown.

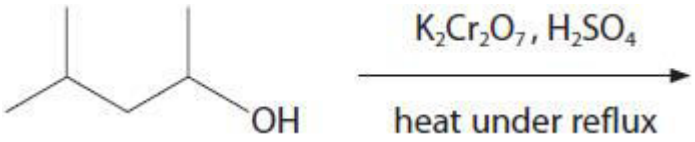
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(3)

Reaction **1**

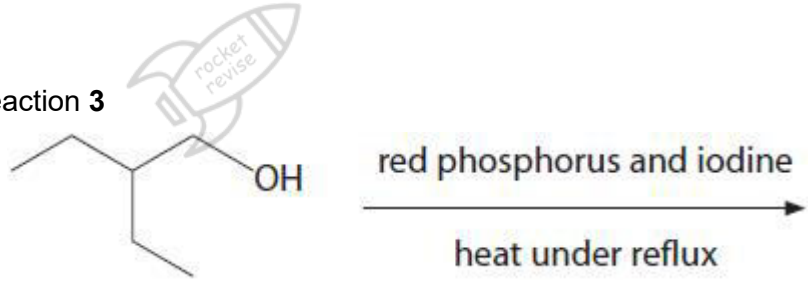


Reaction **2**



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Reaction **3**



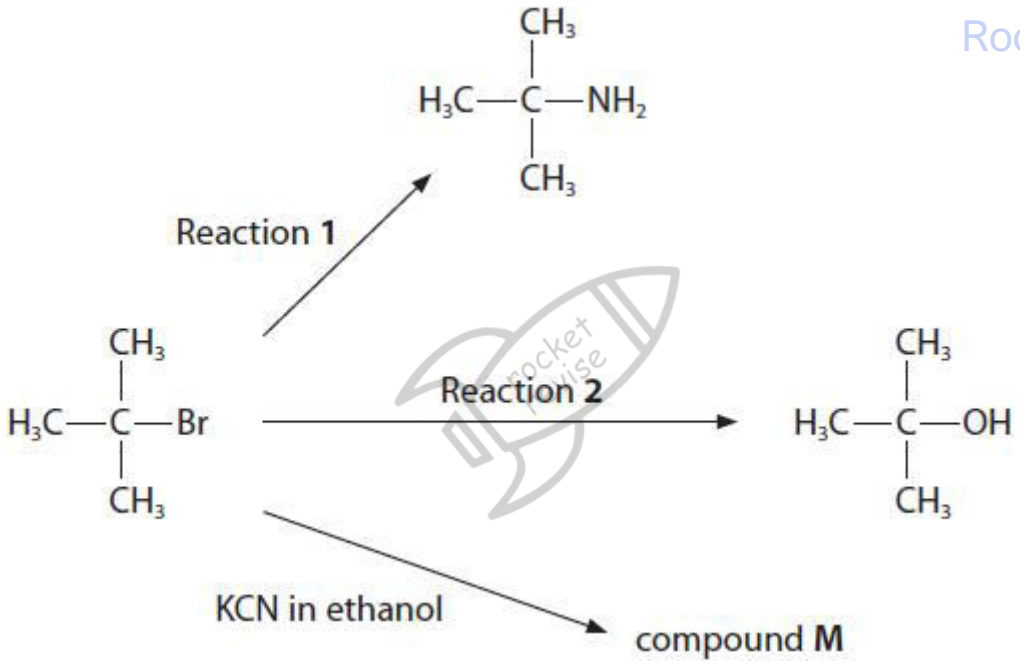
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(Total for question = 12 marks)

Q23.

Some reactions of a halogenoalkane are summarised in the diagram.
All the reactions are of the same type and mechanism.



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(a) Name the type and mechanism of these reactions.

(1)

.....

(b) Give the IUPAC name and the structure of compound **M**.

(2)

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.....

(c) State the reagent and conditions required for Reaction **1**.

(2)

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.....

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(d) An incomplete mechanism for Reaction **2** is shown.

(i) Complete the mechanism by adding curly arrows, and relevant dipoles and lone pairs.

(2)

(ii) Describe how intermediate **J** is formed in the first step, naming the process that occurs.

(2)

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(iii) State the shape of intermediate **J**. Justify your answer.

(2)

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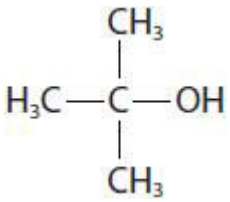
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(iv) The product of Reaction **2** is shown.



This compound has three isomers that are also alcohols.
Draw the structures of these isomers.

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(2)

isomer 1	isomer 2	isomer 3

(e) The reaction of halogenoalkanes with hydroxide ions is often carried out using sodium hydroxide or potassium hydroxide.

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An alternative is 'moist silver(I) oxide', which is prepared by adding a few drops of water to solid silver(I) oxide.
Silver(I) oxide is insoluble in water but produces sufficient hydroxide ions for the reaction.

(i) Write the **ionic** equation for the reaction of silver(I) oxide with water. State symbols are not required.

(2)

(ii) Suggest **one** advantage of using silver(I) oxide rather than sodium hydroxide or potassium hydroxide.

(1)

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(Total for question = 16 marks)

Q24.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Tertiary alcohols are used in the manufacture of petrol additives.

(a) Which of these is a tertiary alcohol?

(1)

- ☐ **A** 1–methylcyclopentanol
- ☐ **B** 2–methylcyclopentanol
- ☐ **C** 2–methylbutan–1–ol
- ☐ **D** 3–methylpentan–2–ol

(b) Which reagent reacts with tertiary alcohols?

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(1)

- ☐ **A** acidified aqueous potassium dichromate(VI)
- ☐ **B** bromine water
- ☐ **C** phosphorus(V) chloride
- ☐ **D** sodium carbonate solution

(Total for question = 2 marks)

Q25.

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Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Propan-1-ol may be converted into propene.

$$CH_3CH_2CH_2OH \rightarrow CH_3CH\&\#9552;CH_2 + H_2O$$

(a) Which reagent is used for this reaction?

(1)

- ☒ **A** 50 % sulfuric acid
- ☒ **B** ethanolic potassium hydroxide
- ☒ **C** phosphoric(V) acid
- ☒ **D** red phosphorus

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(b) The reaction is best classified as

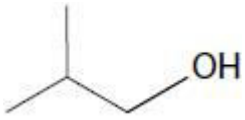
(1)

- ☒ **A** elimination
- ☒ **B** hydrolysis
- ☒ **C** reduction
- ☒ **D** substitution

(Total for question = 2 marks)

Q26.

2-methylpropan-1-ol has the skeletal formula:



(a) 2-methylpropan-1-ol can be converted to 1-bromo-2-methylpropane.

Give the reagents and conditions used for this reaction.

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(2)

Reagents

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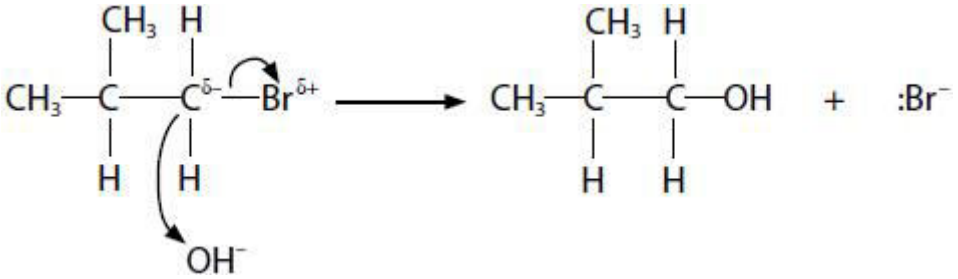
Conditions

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(b) 1-bromo-2-methylpropane can be converted back to 2-methylpropan-1-ol by heating with aqueous alkali. A student suggested the following mechanism for the reaction.

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Identify and correct the three mistakes in the mechanism shown.

(3)

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(c) 1-bromo-2-methylpropane can be converted to 2-methylpropene.

Give the reagents and conditions used for this reaction.

(2)

Reagents

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Conditions

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(Total for question = 7 marks)

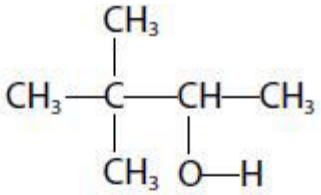
Q27.

Which of the following compounds could be oxidised to a carboxylic acid by refluxing with potassium dichromate(VI) and dilute sulfuric acid?

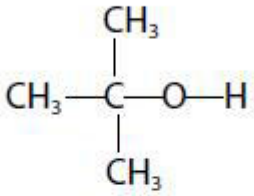
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☐ **A**

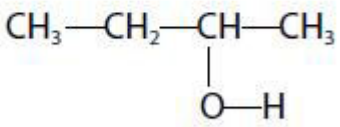


☐ **B**

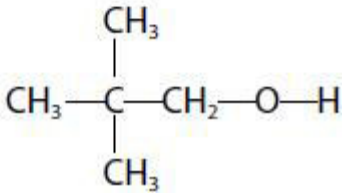


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☐ **C**



☐ **D**

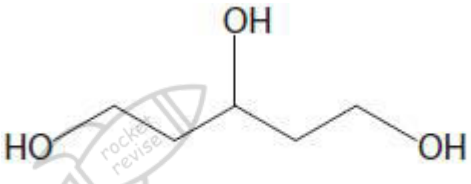


(Total for question = 1 mark)

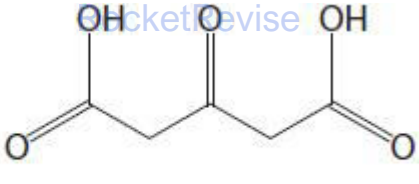
Q28.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What amount, in mol, of [O] is needed to convert 1 mol of pentane-1,3,5-triol into 3-oxopentanedioic acid and water?



pentane-1,3,5-triol



3-oxopentanedioic acid

☐ **A** 2
☐ **B** 3
☐ **C** 4
☐ **D** 5

(Total for question = 1 mark)

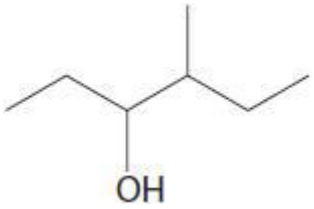
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Q29.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

What is the name of the alcohol shown?



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- ☐ **A** 4-ethyl-4-methylbutan-3-ol
- ☐ **B** 4-methylhexan-3-ol
- ☐ **C** 3-methylhexan-3-ol
- ☐ **D** 2-ethylpentan-3-ol

(Total for question = 1 mark)

Q30.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is a correct equation for the **incomplete** combustion of butan-1-ol?

- ☐ **A** $CH_3CH_2CH_2CH_2OH + 6O_2 \rightarrow 4CO_2 + 5H_2O$
- ☐ **B** $CH_3CH_2CH_2CH_2OH + 6\frac{1}{2}O_2 \rightarrow 4CO_2 + 5H_2O$
- ☐ **C** $CH_3CH_2CH_2CH_2OH + 4O_2 \rightarrow 4CO + 5H_2O$
- ☐ **D** $CH_3CH_2CH_2CH_2OH + 4\frac{1}{2}O_2 \rightarrow 4CO + 5H_2O$

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(Total for question = 1 mark)

Q31.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Which reagent(s) could be used to produce a chloroalkane from a **tertiary** alcohol?

1 PCl_5
2 concentrated HCl
3 concentrated H_2SO_4 and KCl

- ☐ **A** **1, 2** and **3**
- ☐ **B** **1** and **3** only
- ☐ **C** **2** and **3** only
- ☐ **D** **1** only

(Total for question = 1 mark)

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Q32.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which compound needs exactly nine moles of oxygen (O_2) for complete combustion of one mole of the compound?

- ☐ **A** C_5H_{12}
- ☐ **B** $C_6H_{12}O$
- ☐ **C** C_6H_{14}
- ☐ **D** $C_6H_{14}O$

(Total for question = 1 mark)

Q33.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the total number of structural isomers for cyclic alcohols with the molecular formula C_4H_8O?

- ☐ **A** 2
- ☐ **B** 3
- ☐ **C** 4
- ☐ **D** 5

(Total for question = 1 mark)

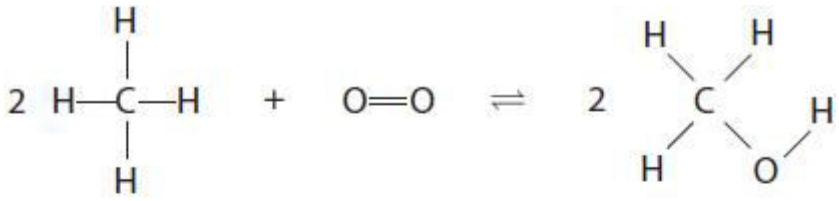
Q34.

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Methanol has been considered as an alternative fuel for many years. Bacteria that metabolise methane could be used for the large-scale production of the alcohol.

(a) Calculate the enthalpy change per mole of methanol produced, using the bond enthalpies shown.

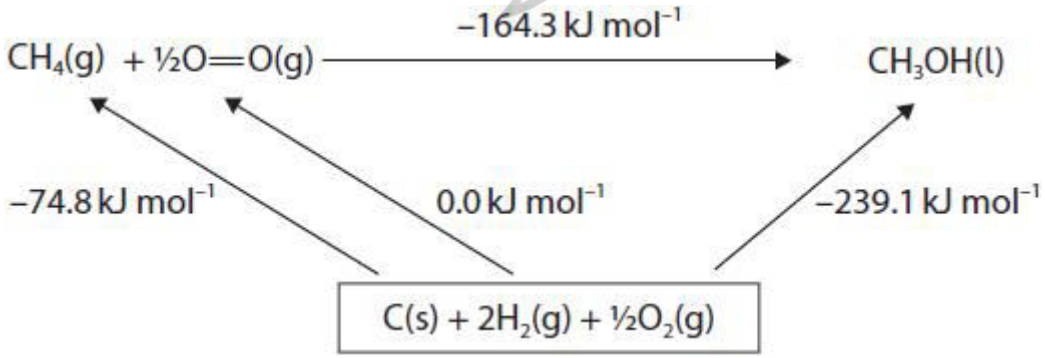


Bond	Enthalpy / kJ mol^{-1}
O═O	498
C—H	413
C—O	336
O—H	464

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(3)

(b) A Hess cycle for the reaction is shown.



State why the standard enthalpy change of formation of oxygen is 0.0 kJ mol^{-1} .

(1)

.....

.....

(c) The standard enthalpy change of reaction to form methanol from methane calculated in part (a) is less accurate than the value given in the Hess cycle in part (b).

Give **two** reasons for this difference.

(2)

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(d) One problem when using bacteria to convert methane to methanol is that the methanol is quickly converted into methanal.

(i) State the type of reaction taking place as methanol is converted into methanal.

(1)

.....

(ii) Give the reagents and conditions that are needed for this reaction in the laboratory. Include in your answer any colour change seen.

(4)

Reagents .....

.....

Conditions .....

Colour change .....

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(e) In industry, methanol can be synthesised in a two-step process.

The first step produces CO, CO_2 and H_2 in two reactions as shown.

$$CH_4 + \tfrac{1}{2}O_2 \rightleftharpoons CH_3OH \qquad \Delta H = -164.3\,kJ\,mol^{-1}$$

This step requires a temperature of 850°C and a pressure of 2500 kPa.
The second step requires a pressure of 8000 kPa but does not require heating as the forward reactions are exothermic.

$$CO + 2H_2 \rightleftharpoons CH_3OH \qquad CO_2 + 3H_2 \rightleftharpoons CH_3OH + H_2O$$

This leads to a yield of around 5 %. Unreacted gases are recycled.
Scientists would prefer a one-step synthesis using a catalyst.

(i) Suggest **two** possible advantages, other than increasing the yield, of a one-step synthesis reaction.

(2)

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(ii) Many different catalysts have been used, including MoO_3, WO_3 and Re.
State **two** characteristics that all these catalysts have in common.

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(2)

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(iii) A one-step process is being developed.
This is carried out at 101 kPa and 800 °C.

$$CH_4 + \tfrac{1}{2}O_2 \rightleftharpoons CH_3OH \qquad \Delta H = -164.3\,kJ\,mol^{-1}$$

Deduce how the reaction conditions could be changed to increase the yield, explaining why such changes may not be economical.

(5)

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(Total for question = 20 marks)

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